

case seems more reasonable from Figure 2.5.2, which shows the function in a small neighborhood of $x = 1$.

A mathematical proof of the fact that the function is arbitrarily often differentiable falls outside the scope of this book. $\square$

The result in Example 2.4.12 can be formulated differently. In fact the example exhibits a function f for which the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ converges for all $x \in \mathbb{R}$, but with a sum which does not equal $f(x)$ for all $x \in \mathbb{R}$.

We notice that it is known exactly which functions can be represented via a power series: it is the class of analytic functions (i.e., the functions that are complex differentiable). More details on this can be found in any standard textbook on complex analysis, e.g. [7].

2.5 General infinite sums of functions

As we have seen, a power series consists of infinite sums of terms $a_n x^n$, $n \in \mathbb{N}$. Since not all functions have a representation of this type, it is natural also to consider infinite sums of other types of “simple” functions. **More generally, we will consider a family of functions $f_0, f_1, f_2, \dots$ with the same domain of definition I , and attempt to define the function**

$$f(x) = f_0(x) + f_1(x) + f_2(x) + \cdots + f_n(x) + \cdots = \sum_{n=0}^{\infty} f_n(x). \quad (2.23)$$

Here it is of course important to know for which values of x the expression for $f(x)$ makes sense: this is the case exactly for the $x \in I$ for which *the series of numbers* $\sum_{n=0}^{\infty} f_n(x)$ is convergent. The function f is called *the sum function* associated to $f_0, f_1, \dots$. As before we will call the expression

$$S_N(x) = f_0(x) + f_1(x) + \cdots + f_N(x)$$

the N th partial sum of $\sum_{n=0}^{\infty} f_n(x)$; it is now a function depending on x .

In practice, one can only deal with a representation of the type (2.23) if the functions f_n have some kind of common structure. In the next chapters we will consider infinite series where the functions f_n are trigonometric functions or elements from a wavelet system; both types of systems satisfy this requirement.

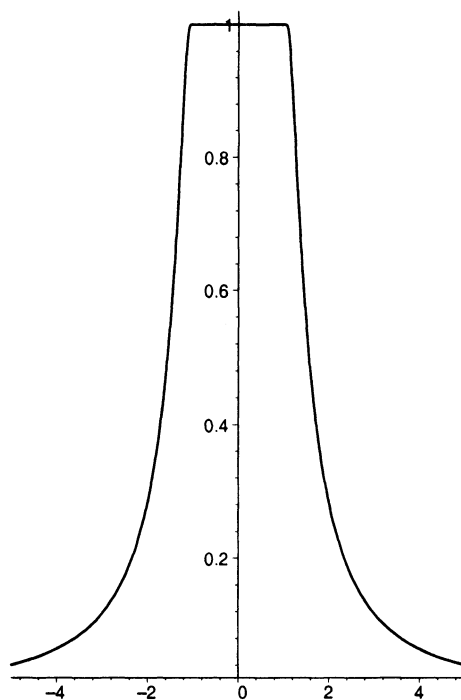


Figure 2.5.1 The function in (2.22), shown on the interval $[-5, 5]$.

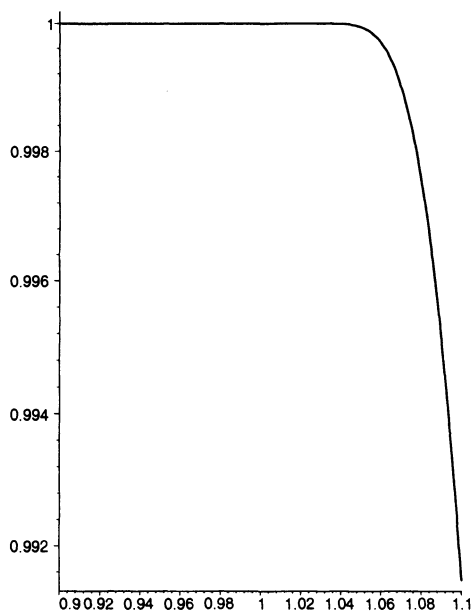


Figure 2.5.2 The function in (2.22), shown on the interval $[0.9, 1.1]$. Notice the unit on the second axis!

We emphasize once more that care is needed while working with infinite sums: many of the results we use for finite sums can not be generalized to infinite sums. For example, we know that any finite sum of continuous functions is continuous; the next example shows that the sum of infinitely many continuous functions does *not* need to be continuous. This demonstrates clearly that we can not just take the known rules for finite sums and use them on infinite sums.

Example 2.5.3 Consider the series

$$\sum_{n=0}^{\infty} x(1-x^2)^n.$$

We wish to find the values of $x \in \mathbb{R}$ for which the series is convergent, and to determine the sum function.

For $x = 0$ the series is convergent with sum 0. For a fixed $x \neq 0$ we can view the series as a geometric series with quotient $1 - x^2$, which is multiplied with the number x ; by Theorem 2.3.3, the series is convergent for $|1 - x^2| < 1$, i.e., for $0 < |x| < \sqrt{2}$, and divergent for $|x| \geq \sqrt{2}$. For $0 < |x| < \sqrt{2}$ we obtain the sum

$$\sum_{n=0}^{\infty} x(1-x^2)^n = x \frac{1}{1-(1-x^2)} = x \frac{1}{x^2} = \frac{1}{x}.$$

Thus the sum function is

$$f(x) = \sum_{n=0}^{\infty} x(1-x^2)^n = \begin{cases} \frac{1}{x} & \text{for } 0 < |x| < \sqrt{2}, \\ 0 & \text{for } x = 0. \end{cases}$$

We see that the sum function has a strong discontinuity at 0 even though all functions $x(1-x^2)^n$ are continuous! Figure 2.5.5 shows the sum function; compare with the partial sums for $N = 5$ and for $N = 50$, shown on Figure 2.5.6. $\square$

The difference between finite and infinite sums can be used to make surprising and non-intuitive constructions. For example, one can construct a function $f: \mathbb{R} \rightarrow \mathbb{R}$, which is continuous, but nowhere differentiable:

Example 2.5.4 Given constants $A > 1, B \in]0, 1[$, for which $AB \geq 1$, we attempt to consider the function

$$f(x) = \sum_{n=1}^{\infty} B^n \cos(A^n x), \quad x \in \mathbb{R}.$$

Since $|B^n \cos(A^n x)| \leq B^n$ and $B \in]0, 1[$, Theorem 2.1.4 combined with Theorem 2.3.3 show that the series defining f is actually convergent. A result stated formally in the next section, Theorem 2.6.5, tells us that f is continuous.

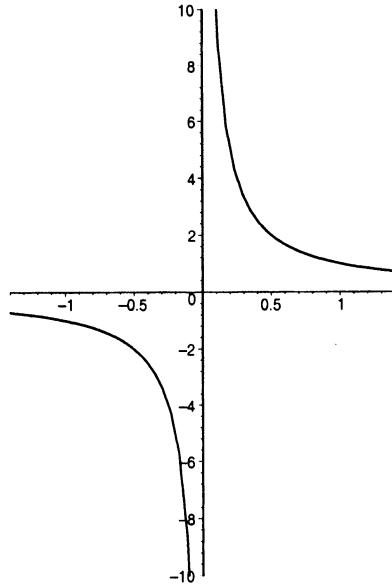


Figure 2.5.5 The function $f(x) = \sum_{n=0}^{\infty} x(1-x^2)^n$, $x \in [-\sqrt{2}, \sqrt{2}]$.

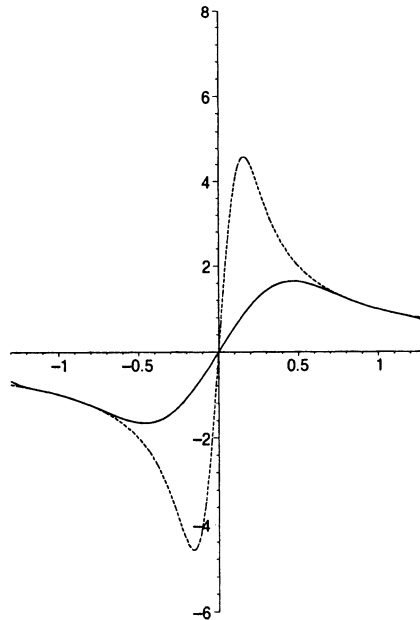


Figure 2.5.6 The partial sums $S_5(x) = \sum_{n=0}^5 x(1-x^2)^n$ (unbroken line) and $S_{50}(x) = \sum_{n=0}^{50} x(1-x^2)^n$ (dotted). The more terms we include in the partial sum, the more the graph looks like the graph of the sum function in Figure 2.5.5. However, all partial sums “turn off” near the point $(0,0)$, where all the graphs go through.

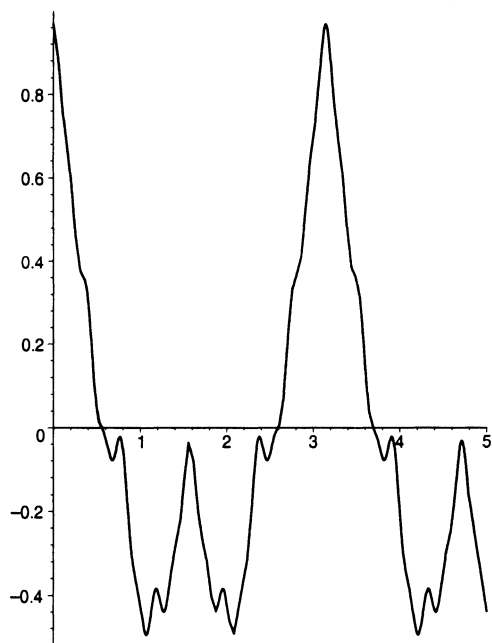


Figure 2.5.7 The partial sum $S_5(x) = \sum_{n=1}^5 \frac{\cos(2^n x)}{2^n}$ of the function in (2.24).

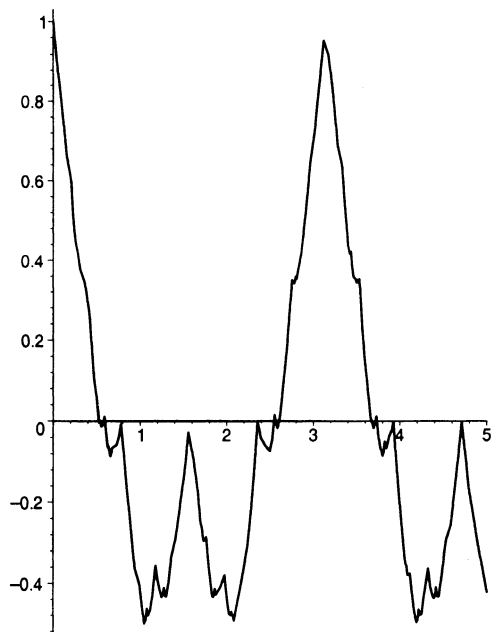


Figure 2.5.8 The partial sum $S_{50}(x) = \sum_{n=1}^{50} \frac{\cos(2^n x)}{2^n}$ of the function in (2.24).

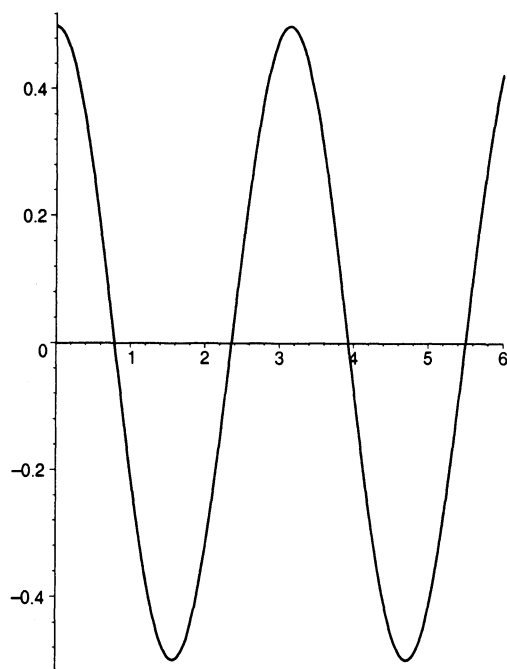


Figure 2.5.9 The function $f_1(x) = \frac{1}{2} \cos(2x)$.

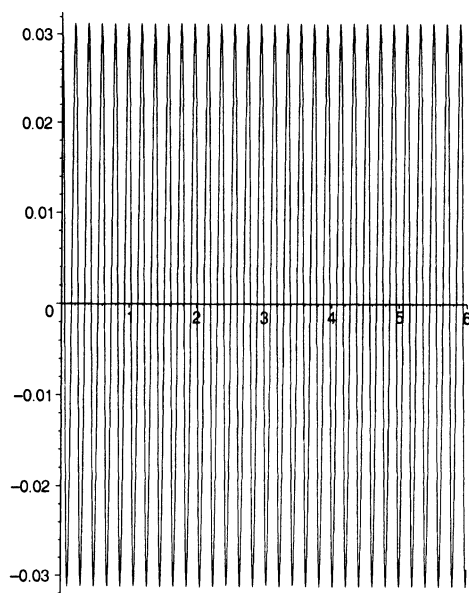


Figure 2.5.10 The function $f_5(x) = \frac{1}{32} \cos(32x)$. Observe the units on the axes!

Assuming that A is an odd integer and that the product AB is sufficiently large, Weierstrass proved in 1887 that the function f is nowhere differentiable. With only classical analysis at hand, this is rather involved; in Example 5.6.4 we return to this function and indicate how a short and elegant proof can be given via wavelet-inspired methods. At this moment, we only aim at an intuitive understanding of the non-differentiability. Let us consider the special case

$$f(x) = \sum_{n=1}^{\infty} \frac{\cos(2^n x)}{2^n}. \quad (2.24)$$

The function f in (2.24) has the form (2.23) with

$$f_n(x) = \frac{\cos(2^n x)}{2^n}.$$

In a rather intuitive sense, the first terms in the series deliver the numerically largest contribution: we see that

$$|f_n(x)| \leq \frac{1}{2^n} \rightarrow 0 \text{ for } n \rightarrow \infty.$$

Due to the oscillatory behavior of the functions f_n , this intuitive statement is not completely true pointwise, but if we adapt the point of view anyway, we can consider the function f as a composition of the harmonic function $f_1(x) = 1/2 \cos(2x)$, see Figure 2.5.9, and the infinite sum of higher harmonics

$$\frac{1}{4} \cos(4x) + \frac{1}{8} \cos(8x) + \cdots.$$

The partial sums S_5 and S_{50} are shown in Figures 2.5.7–2.5.8. The function f_1 is shown in Figure 2.5.9, and Figure 2.5.10 shows the fifth term

$$f_5(x) = \frac{1}{32} \cos(32x);$$

already this term is very small, but it oscillates strongly. This phenomenon becomes more noticeable for larger values of n , i.e., we obtain oscillations with very high frequencies. The consequence is that the sum function f oscillates so much around each point x that it is not differentiable.

In the context of Fourier analysis we will see another indication of the relationship between differentiability and the content of oscillations; see Section 3.7. $\square$

2.6 Uniform convergence

So far, our definition of convergence of an infinite series of functions has been *pointwise*: in fact, (2.23) merely means that if we fix an arbitrary x in

the domain of f , the partial sums of the series converge to $f(x)$. However, as we will explain now, this type of convergence is not sufficient in all situations.

As already mentioned, the motivation behind (2.23) is to write a complicated signal f as an infinite sum of simpler functions. However, since a computer can only deal with finite sums, for applications of this result in practice we always have to replace the infinite sum in (2.23) by a *finite sum*, i.e., by a certain partial sum. In other words, we need to be able to find $N \in \mathbb{N}$ such that

$$|f(x) - S_N(x)|$$

is sufficiently small, *simultaneously for all x in the considered interval*. That is, in the terminology used in Section 1.1, we need that the partial sums deliver approximations of f in the uniform sense. As the following example demonstrates, this is not always the case.

Example 2.6.1 Consider the function

$$f(x) = \frac{1}{1-x}, \quad x \in]-1, 1[.$$

Then, as we saw in Theorem 2.3.3,

$$f(x) = \sum_{n=0}^{\infty} x^n, \quad (2.25)$$

understood in the sense that the partial sums $S_N(x)$ of the right-hand side of (2.25) converges to $f(x)$ for any $x \in]-1, 1[$. However, if we fix any $N \in \mathbb{N}$, we have

$$\begin{aligned} \left| f(x) - \sum_{n=0}^N x^n \right| &= \left| \sum_{n=N+1}^{\infty} x^n \right| \\ &= \frac{|x|^{N+1}}{1-x}. \end{aligned}$$

No matter how large we choose $N \in \mathbb{N}$, there exists an $x \in]-1, 1[$ for which this difference is as large as we would like to ask for (see Figures 2.3.1–2.3.2). Thus, we are not able to obtain a good approximation simultaneously for all $x \in]-1, 1[$; this is a serious shortcoming for practical applications.

Note that the above problem disappears if we only consider the function f on any interval of the type $[-r, r]$ for some $r \in]0, 1[$; in this case, for any $x \in [-r, r]$ we have

$$\begin{aligned} \left| f(x) - \sum_{n=0}^N x^n \right| &= \frac{|x|^{N+1}}{1-x} \\ &\leq \frac{r^{N+1}}{1-r}. \end{aligned}$$

Since

$$\frac{r^{N+1}}{1-r} \rightarrow 0 \text{ as } N \rightarrow \infty,$$

we can thus make $\left| f(x) - \sum_{n=0}^N x^n \right|$ as small as we want, simultaneously for all $x \in [-r, r]$, by choosing $N \in \mathbb{N}$ sufficiently large. $\square$

Example 2.6.1 suggests that we introduce the following type of convergence.

Definition 2.6.2 *Given functions $f_1, f_2, \dots, f_n, \dots$ defined on an interval I , assume that the function*

$$f(x) = \sum_{n=1}^{\infty} f_n(x), \quad x \in I,$$

is well defined. Then we say that $\sum_{n=1}^{\infty} f_n$ converges uniformly to f on the interval I if for each $\epsilon > 0$ we can find an $N_0 \in \mathbb{N}$ such that

$$\left| f(x) - \sum_{n=1}^N f_n(x) \right| \leq \epsilon \text{ for all } x \in I \text{ and all } N \geq N_0.$$

Note that an equivalent definition would be to say that $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly to f on the interval I if for each $\epsilon > 0$ we can find $N_0 \in \mathbb{N}$ such that

$$\sup_{x \in I} \left| f(x) - \sum_{n=1}^N f_n(x) \right| \leq \epsilon \text{ for all } N \geq N_0.$$

Let us return to Example 2.6.1:

Example 2.6.3 Formulated in terms of uniform convergence, our result in Example 2.6.1 shows that

- $\sum_{n=0}^{\infty} x^n$ does not converge uniformly to $f(x) = \frac{1}{1-x}$ on $I =]-1, 1[$;
- $\sum_{n=0}^{\infty} x^n$ converges uniformly to $f(x) = \frac{1}{1-x}$ on any interval $I = [-r, r]$, $r \in]0, 1[$.

It turns out that Example 2.6.3 is typical for power series: in general, nothing guarantees that a power series with radius of convergence ρ converges uniformly on $] -\rho, \rho[$. On the other hand we always obtain uniform convergence if we shrink the interval slightly:

Proposition 2.6.4 *Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $\rho > 0$. Then $\sum_{n=0}^{\infty} a_n x^n$ is uniformly convergent on any interval of the form $[-r, r]$, where $r \in]0, \rho[$.*

Proof: Given any $x \in [-r, r]$,

$$\begin{aligned} \left| \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^N a_n x^n \right| &= \left| \sum_{n=N+1}^{\infty} a_n x^n \right| \\ &\leq \sum_{n=N+1}^{\infty} |a_n x^n| \\ &\leq \sum_{n=N+1}^{\infty} |a_n r^n|; \end{aligned}$$

thus

$$\sup_{x \in [-r, r]} \left| \sum_{n=0}^{\infty} a_n x^n - \sum_{n=0}^N a_n x^n \right| \leq \sum_{n=N+1}^{\infty} |a_n r^n|. \quad (2.26)$$

By the fact that $\sum_{n=1}^{\infty} a_n r^n$ is absolutely convergent, the quantity on the right-hand side of (2.26) goes to zero as $N \rightarrow \infty$. $\square$

We end this section with a few important results concerning continuity and differentiability of infinite series of functions; the first result below gives conditions for the associated sum function being well defined and continuous.

Theorem 2.6.5 *Assume that the functions $f_1, f_2, \dots$ are defined and continuous on an interval I , and that there exist positive constants $k_1, k_2, \dots$ such that*

- (i) $|f_n(x)| \leq k_n, \forall x \in I, n \in \mathbb{N}$;
- (ii) $\sum_{n=1}^{\infty} k_n$ is convergent.

Then the infinite series $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly; and the function

$$f(x) = \sum_{n=1}^{\infty} f_n(x), \quad x \in I$$

is continuous.

A series $\sum_{n=1}^{\infty} k_n$ satisfying the condition (i) in Theorem 2.6.5 is called a *majorant series* for $\sum_{n=1}^{\infty} f_n(x)$. Theorem 2.6.5 is known in the literature under the name *Weierstrass' M-test*.

If we want the sum function to be differentiable, we need slightly different conditions, stated below; on the theoretical level, the result can be used to prove Theorem 2.4.10.

Theorem 2.6.6 *Assume that the functions $f_1, f_2, \dots$ are defined and differentiable with a continuous derivative on the interval I , and that the*

function $f(x) = \sum_{n=1}^{\infty} f_n(x)$ is well defined on I . Assume also that there exist positive constants $k_1, k_2, \dots$ such that

$$(i) \quad |f'_n(x)| \leq k_n, \quad \forall x \in I, \quad n \in \mathbb{N};$$

$$(ii) \quad \sum_{n=1}^{\infty} k_n \text{ is convergent.}$$

Then f is differentiable on I , and

$$f'(x) = \sum_{n=1}^{\infty} f'_n(x).$$

Finally, we mention that uniform convergence of an infinite series consisting of integrable functions allows us to integrate term-wise:

Proposition 2.6.7 Assume that the functions $f_1, f_2, \dots$ are continuous on the interval I and that $\sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent. Then, the function $x \mapsto \sum_{n=1}^{\infty} f_n(x)$ is continuous on I ; furthermore, for any $a, b \in I$,

$$\int_a^b \sum_{n=1}^{\infty} f_n(x) dx = \sum_{n=1}^{\infty} \int_a^b f_n(x) dx.$$

Proposition 2.6.7 has an important consequence for integration of power series:

Corollary 2.6.8 Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $\rho > 0$. Then, for any $b \in]-\rho, \rho[$,

$$\int_0^b \sum_{n=0}^{\infty} a_n x^n dx = \sum_{n=0}^{\infty} \frac{a_n}{n+1} b^{n+1}.$$

2.7 Signal transmission

Modern technology often requires that information can be sent from one place to another; one speaks about *signal transmission*. It occurs, e.g., in connection with wireless communication, the internet, computer graphics, or transfer of data from CD-ROM to computer.

All types of signal transmission are based on transmission of a series of numbers. The first step is to convert the given information (called the *signal*) to a series of numbers, and this is where the question of having a series representation comes in: if we know that a signal is given as a function f which has a power series representation

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad (2.27)$$

then all information about the function f is stored in the coefficients $\{a_n\}_{n=0}^{\infty}$. In other words: if we know the coefficients $\{a_n\}_{n=0}^{\infty}$, then we can immediately use (2.27) to find out which signal we are dealing with.

Let us explain how this can be used in signal transmission. Assume for example that a sender $\mathcal{S}$ wishes to send a picture to a receiver $\mathcal{R}$. For simplification, we assume that the picture is given as a graph of a function f , and that f can be represented as a power series. Then the transmission can be done in the following way:

- $\mathcal{S}$ finds the coefficients $a_0, a_1, \dots$ such that $f(x) = \sum_{n=0}^{\infty} a_n x^n$;
- $\mathcal{S}$ sends the coefficients $a_0, a_1, \dots$;
- $\mathcal{R}$ receives the coefficients $a_0, a_1, \dots$;
- $\mathcal{R}$ reconstructs the signal by multiplying the coefficients a_n by x^n and forming the infinite series $f(x) = \sum_{n=0}^{\infty} a_n x^n$.

In practice there are a few more steps not described here. For example, $\mathcal{S}$ can not send an infinite sequence of numbers $a_0, a_1, \dots$: it is possible to send only a finite sequence of numbers $a_0, a_1, \dots, a_N$. The choice of the numbers to send needs to be done with care: we want that the signal we actually transmit, i.e.,

$$f^\#(x) = a_0 + a_1 x + \dots + a_N x^N = \sum_{n=0}^N a_n x^n,$$

looks and behaves like the original signal

$$f(x) = a_0 + a_1 x + \dots + a_N x^N + \dots = \sum_{n=0}^{\infty} a_n x^n.$$

This usually forces N to be large. Another issue is that $\mathcal{S}$ can only send the numbers $a_0, \dots, a_N$ with a certain precision, for example, 50 digits. So in reality $\mathcal{S}$ will send some numbers

$$\tilde{a}_0 \simeq a_0, \quad \tilde{a}_1 \simeq a_1, \quad \tilde{a}_2 \simeq a_2, \quad \dots, \quad \tilde{a}_N \simeq a_N,$$

and $\mathcal{R}$ receives coefficients corresponding to the signal

$$\tilde{f}(x) = \sum_{n=0}^N \tilde{a}_n x^n.$$

The fact that the numbers a_n can only be sent with a certain precision is well known in engineering; one speaks about *quantification*.

The process above needs to be made such that

$$\tilde{f}(x) \simeq f(x)$$

— it does not help if $\mathcal{S}$ sends a picture of an elephant and $\mathcal{R}$ receives something that looks like a giraffe! Sometimes it is unsuitable just to send

the *first* $N + 1$ coefficients $a_0, \dots, a_N$: it will be much more reasonable to send the $N + 1$ most important coefficients, which usually means the (numerically) *largest* coefficients. In this case one also has to send information about which powers x^n the coefficients that are sent belong to; we speak about *coding* of the information.

The idea of sending the $N + 1$ largest coefficients instead of just the first $N + 1$ coefficients can be seen as the first step towards *best N -term approximation*, a subject to which we return in Section 3.8 and Section 5.7. Observe that in the technical sense, to send the largest coefficients first usually corresponds to *reordering* the terms in the given infinite series; as we have seen in Section 2.1 this can lead to further complications if the series does not converge unconditionally, so it has to be done with care.

We also note that to send only a finite number of numbers $a_0, a_1, \dots, a_N$, rather than the full sequence, corresponds to a *compression*, in the sense that a part of the information determining the signal is thrown away. In the situation discussed here, this type of compression can not be avoided: we need to have a finite sequence in order to be able to transmit. Usually the word “compression” appears in a slightly different sense, namely, that we want the information given in a finite sequence to be represented efficiently using a shorter sequence. This is a very important issue when dealing with large data sets; we will return to it in Section 4.2.

Example 2.7.1 Assume that $\mathcal{S}$ wishes to send the function

$$f(x) = \sin x, \quad x \in [0, 3]$$

to $\mathcal{R}$. In Example 2.4.4 we showed that

$$\sin x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^{2n+1}}{(2n+1)!} = 0 + x + 0 \cdot x^2 - \frac{1}{6}x^3 + 0 \cdot x^4 + \frac{1}{120}x^5 + \dots$$

Since $\frac{1}{6} \approx 0.1666$, $\frac{1}{120} \approx 0.0083$, $\mathcal{S}$ can choose to send the numbers

$$0; 1; 0; -0.1666; 0; 0.0083,$$

according to which $\mathcal{R}$ will reconstruct the function as

$$\tilde{f}(x) = x - 0.1666x^3 + 0.0083x^5 \approx f(x).$$

Figure 2.7.2 shows that the reconstructed signal is close to the original signal when $x \in [0, 2]$, but then the approximation starts to deviate from the signal. A better reconstruction can be reached if we also send the coefficient belonging to x^7 , namely $-\frac{1}{7!} \approx -0.000198$; then

$$\tilde{f}(x) = x - 0.1666x^3 + 0.0083x^5 - 0.000198x^7,$$

see Figure 2.7.3. □

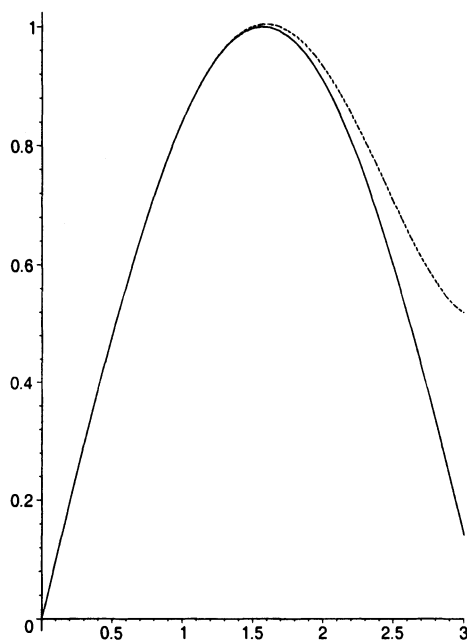


Figure 2.7.2 The functions $f(x) = \sin x$ (unbroken line) and $\tilde{f}(x) = x - 0.1666x^3 + 0.0083x^5$.

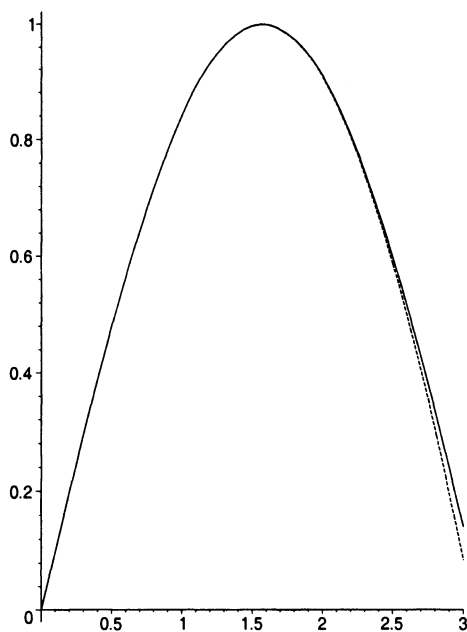


Figure 2.7.3 The functions $f(x) = \sin x$ (unbroken line) and $\tilde{f}(x) = x - 0.1666x^3 + 0.0083x^5 - 0.000198x^7$.

In practice, transmission of pictures is done in a different way than described here, see Section 4.3. One of the problems with the principle above is that not all functions have a power series representation. Thus, in many situations we are forced to work with other types of series representations, a topic to which we return in the next chapters.

2.8 Exercises

2.1 Which of the series below are convergent?

- (i) $\sum_{n=1}^{\infty} \frac{\sin n + \cos n}{n^2}$
- (ii) $\sum_{n=1}^{\infty} \cos(n\pi) \frac{2}{n+5}$
- (iii) $\sum_{n=1}^{\infty} (-2)^n \frac{1}{n^2+7}$
- (iv) $\sum_{n=2}^{\infty} \frac{1}{\log n}$
- (v) $\sum_{n=2}^{\infty} \frac{(-1)^n}{\log n}$.

2.2 Prove the inequalities below via the integral test:

$$\frac{\pi}{4} < \sum_{n=1}^{\infty} \frac{1}{n^2+1} < \frac{\pi}{4} + \frac{1}{2},$$

$$\frac{1}{8} < \sum_{n=2}^{\infty} \frac{1}{n^3} < \frac{1}{4}.$$

2.3 Find an approximative value for $\sum_{n=1}^{\infty} \frac{1}{n^4}$, with an error of maximally 0.02.

2.4 Prove that

$$\sum_{n=N+1}^{\infty} \frac{1}{(2n+1)^2} \leq \left(\frac{1}{4N+6} + \frac{1}{(2N+3)^2} \right), \quad \forall N \in \mathbb{N}.$$

2.5 Prove that the series

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{4n-3}} + \frac{1}{\sqrt{4n-1}} - \frac{1}{\sqrt{2n}} \right)$$

is divergent (hint: look at $\sqrt{n}a_n$ as $n \rightarrow \infty$).

Looking at a_n for the first few values of n , we observe that $\sum_{n=1}^{\infty} a_n$ consists of terms of the form $\pm \frac{1}{\sqrt{n}}$, with the sign plus when n is odd, and the sign minus when n is even. According to Proposition 2.2.4, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ is convergent. Explain why these results do not contradict each other.

2.6 Compare the estimates for $\int_0^1 e^{-x^2} dx$ in

- Example 2.4.3, using the partial sums of (2.16) corresponding to $N = 5$ and $N = 8$;
- Exercise 1.3.

2.7 Find the radius of convergence for the power series

$$\sum_{n=1}^{\infty} nx^n,$$

and find an expression for the sum in terms of the standard functions.

2.8 Find the radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{n+2}{n+1} x^{n+1},$$

and find an expression for the sum in terms of the standard functions.

2.9 (i) Find the radius of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{n+1}{n \cdot n!} x^n.$$

Let $f(x)$ denote the sum of this series.

(ii) Find a power series representation for the function

$$h(x) = \int_0^x t f'(t) dt, \quad x \in \mathbb{R}.$$

(iii) Use the power series for e^x to find an expression for $h(x)$ in terms of the standard functions.

2.10 Prove that

$$f(x) := \sum_{n=1}^{\infty} \frac{3 \cos nx - \sin^2 nx}{3^n - 1}, \quad x \in \mathbb{R},$$

defines a continuous function.

2.11 (i) Prove that the function

$$f(x) := \sum_{n=0}^{\infty} \frac{1}{3^n} \cos(2^n x), \quad x \in \mathbb{R},$$

is well defined.

(ii) Prove that f is continuous.

(iii) Prove that f is differentiable, and find $f'(0)$.

2.12 Find the values of $x \in \mathbb{R}$ for which

$$\sum_{n=0}^{\infty} \frac{1}{(2+x^2)^n}$$

is convergent, and express the sum via the standard functions.

2.13 Prove that

$$\sum_{n=k+1}^{\infty} \frac{1}{(2n-1)^2} \leq \frac{1}{4k+2} + \frac{1}{(2k+1)^2}, \quad \forall k \in \mathbb{N}.$$

2.14 Find a polynomial $P(x)$ such that

$$|\sin x - P(x)| \leq 0.05 \text{ for all } x \in [0, 3].$$

2.15 Assume that we want to transmit a function having a power series expansion. Explain why uniform convergence of the power series is relevant in this context.